



Chitkara

PAPER OF CSE-2025 - PYTHON TEST 2 EXAM

Q1. Puzzle Piece Acquisition Rate

In an adventure game, the number of puzzle pieces acquired per stage follows a pattern. The first stage yields 1 piece. Each subsequent stage yields double the pieces of the stage before it. A challenging boss fight occurs every 5th stage, and after defeating it, the player's next piece acquisition rate resets to 1 piece.

Task: Calculate the total number of puzzle pieces acquired after S stages.

Input Format:

The total number of stages, S, is given as an integer.

Output Format:

Print the calculated total number of puzzle pieces acquired.

Sample Input 1:

6

Sample Output 1:

18

Explanation:

Pieces acquired sequence by stage:

Stage 1 → 1

Stage 2 → 2

Stage 3 → 4

Stage 4 → 8

Stage 5 → 1 (rate reset)

Stage 6 → 2

Total = 18 pieces

ANSWER:

Test case 1 :

Input:

6

Output:

18

Test case 2 :

Input:

5

Output:

16

Test case 3 :

Input:

1

Output:

1

Test case 4 :

Input:

7

Output:

22

Test case 5 :

Input:

0

Output:

0

Test case 6 :

Input:

13

Output:

46

Q2. Lighthouse Beacon Brightness

A newly automated lighthouse beacon has a complex brightness pattern. The initial brightness in the first flash is 1 lumen. In every subsequent flash, the brightness doubles. For cooling purposes, the system initiates an internal dampening every 5th flash, which resets the brightness of the next flash back to 1 lumen.

Task: Determine the aggregate brightness emitted by the beacon across F total flashes.

Input Format:

The total number of flashes, F, is given as an integer.

Output Format:

Print the calculated aggregate brightness emitted by the beacon.

Sample Input 1:

7

Sample Output 1:

22

Explanation:

Brightness sequence by flash:

1st flash → 1 lumen

2nd flash → 2 lumen

3rd flash → 4 lumen

4th flash → 8 lumen

5th flash → 1 lumen (crash/reset)

6th flash → 2 lumen

7th flash → 4 lumen

Total = 22 lumen

ANSWER:Test case 1 :

Input:

7

Output:

22

Test case 2 :

Input:

5

Output:

16

Test case 3 :

Input:

1

Output:

1

Test case 4 :

Input:

7

Output:

22

Test case 5 :

Input:

0

Output:

0

Test case 6 :

Input:

13

Output:

46

Q3. Stock Market Volatility Index

A fictional stock index shows an increase in volatility each week. The initial increase is 1 point. In each following week, the increase doubles. To simulate market correction, the increase is reset to 1 point every 5th week.

Task: Calculate the total weekly index point increases over W total weeks.

Input Format:

A single integer value W , representing total weeks.

Output Format:

Print the calculated overall total weekly index point increases.

Sample Input 1:

6

Sample Output 1:

18

Explanation:

Increase sequence by week:

1st Week $\rightarrow$ 1

2nd Week $\rightarrow$ 2

3rd Week $\rightarrow$ 4

4th Week $\rightarrow$ 8

5th Week→ 1 (reset)

6th Week→ 2

total=18

ANSWER:

Test case 1 :

Input:

6

Output:

18

Test case 2 :

Input:

5

Output:

16

Test case 3 :

Input:

1

Output:

1

Test case 4 :

Input:

7

Output:

22

Test case 5 :

Input:

0

Output:

0

Test case 6 :

Input:

13

Output:

46

Q4. Movie Reel Projection Length

A classic movie theater uses film reels. Each **reel** has a fixed viewing time equivalent to **1000 seconds**. The total number of reels needed for a list of movies is calculated. If a movie's run time is not perfectly divisible by 1000 seconds, an **extra reel** is dedicated to the remaining footage.

Task: Calculate the **total number of reels** required for a list of movie run times given in seconds.

Input Format:

A list of integers S, representing movie run times (in seconds).

Output Format:

Output the total reels required.

Sample Input:

[6000, 1, 1000, 1001]

Sample Output:

10

Explanation:

For each run time, reels needed are :

6000 sec → 6 reels

1 sec → 1 reel

1000 sec → 1 reels

1001 sec → 2 reels

Total = 6 + 1 + 1 + 2 = 10

ANSWER:

Test case 1 :

Input:

[6000, 1, 1000, 1001]

Output:

10

Test case 2 :

Input:

[1000, 2000, 3000]

Output:

6

Test case 3 :

Input:

[999, 1, 1001]

Output:

4

Test case 4 :

Input:

[5000]

Output:

5

Test case 5 :

Input:

[]

Output:

0

Test case 6 :

Input:

[0, 200, 400]

Output:

2

Q5. Digital Transaction Fee Calculation

A digital finance platform calculates a transaction fee based on the amount transferred. For every **1000 units** of currency transferred, a **single fee unit** is charged. If the transferred amount is not perfectly divisible by 1000, an **additional fee unit** is charged for the partial amount.

You must calculate the **total fee units** charged for a list of transaction amounts.

Input Format:

A list of integers T, representing transaction amounts.

Output Format:

Output the total fee units charged.

Sample Input:

[500, 1500, 2500, 3000, 1, 2000]

Sample Output:

12

Explanation:

For each transaction, fee units needed are :

500 units → 1 fee

1500 units → 2 fee

2500 units → 3 fee

3000 units → 3 fee

1 units → 1 fee

2000 units → 2 fee

Total = 1 + 2 + 3 + 3 + 1 + 2 = 12

ANSWER:

Test case 1 :

Input:

[500, 1500, 2500, 3000, 1, 2000]

Output:

12

Test case 2 :

Input:

[1000, 2000, 3000]

Output:

6

Test case 3 :

Input:

[999, 1, 1001]

Output:

4

Test case 4 :

Input:

[5000]

Output:

5

Test case 5 :

Input:

[]

Output:

0

Test case 6 :

Input:

[0, 200, 400]

Output:

Q6. Art Supply Canvas Roll Usage

An art studio buys canvas in large rolls. Each **roll** contains **1000 square feet** of canvas. To fulfill orders, the studio must use the minimum number of rolls. If an order's square footage is not exactly divisible by 1000, an **entire extra roll** is considered used for the remaining amount.

Determine the **total number of canvas rolls** used for a list of project area requirements.

Input Format:

The input is a list of integers A, representing the required canvas areas (in square ft).

Output Format:

Print the total rolls used for given list.

Sample Input:

[999, 1, 2000]

Sample Output:

4

Explanation:

For each area, rolls needed are:

999 sq ft → 1 roll

1 sq ft → 1 roll

2000 sq ft → 2 rolls

Total = 1 + 1 + 2 = 4

ANSWER:

Test case 1 :

Input:

[999, 1, 2000]

Output:

4

Test case 2 :

Input:

[1000, 2000, 3000]

Output:

6

Test case 3 :

Input:

[999, 1, 1001]

Output:

4

Test case 4 :

Input:

[5000]

Output:

5

Test case 5 :

Input:

[]

Output:

0

Test case 6 :

Input:

[0, 200, 400]

Output:

2

Q7. Grandma Secret Recipe Decoder

Grandma wrote her recipes backward.

If recipe contains *, @, /, the page is spoiled.

You must safely decode the message.

Rules:

1. Take a string message as input.
2. If the message has @, #, or \$, raise → "Invalid Symbol Found".
3. Otherwise, just reverse each word and print the decoded message.

Input Format:

One string (encrypted message)

Output Format:

Decoded message or error message.

Sample test case 1 :

Input:

retaw dda dna ksab

Output:

water add and bask

Sample test case 2:**Input:**

retaw fi /dda

Output:

Invalid Symbol Found

ANSWER:Test case 1 :

Input:

retaw dda dna ksab

Output:

water add and bask

Test case 2 :

Input:

@olleh dlrow

Output:

Invalid Symbol Found

Test case 3 :

Input:

ym eman si dipa

Output:

my name is apid

Test case 4 :

Input:

nohtyp si nuf

Output:

python is fun

Test case 5 :

Input:

dlrow#olleh

Output:

Invalid Symbol Found

Test case 6 :

Input:

\$siht si ysae

Output:

Invalid Symbol Found

Q8. Library Spellbook Translation

Ancient spellbooks reverse each word.

Forbidden marks {, }, & indicate dangerous magic.

You must safely decode the message.

Rules:

1. Take a string message as input.
2. If {, }, & or any symbol found then return→ "Invalid Symbol Found".
3. Otherwise, just reverse each word and print the decoded message.

Input Format:

One string (encrypted message)

Output Format:

Decoded message or error message.

Sample test case 1 :

Input:

erif lortnoc lliw uoy

Output:

fire control will you

Sample test case 2:

Input:

erif{lortnoc ew

Output:

Invalid Symbol Found

ANSWER:

Test case 1 :

Input:

erif lortnoc lliw uoy

Output:

fire control will you

Test case 2 :

Input:

@olleh dlrow

Output:

Invalid Symbol Found

Test case 3 :

Input:

ym eman si dipa

Output:

my name is apid

Test case 4 :

Input:

nohtyp si nuf

Output:

python is fun

Test case 5 :

Input:

dlrow#olleh

Output:

Invalid Symbol Found

Test case 6 :

Input:

\$siht si ysae

Output:

Invalid Symbol Found

Q9. Alien Language Interpreter

Aliens talk in reversed words.

Symbols !, %, ~ indicate message distortion.

You must safely decode the message.

Rules:

1. Take a string message as input.
2. If the message has @, #, \$ or any symbol then return → "Invalid Symbol Found".
3. Otherwise, just reverse each word and print the decoded message.

Input Format:

One string (encrypted message)

Output Format:

Decoded message or error message.

Sample test case 1 :**Input:**

era ew sdneirf

Output:

are we friends

Sample test case 2:**Input:**

era%s dneirf

Output:

Invalid Symbol Found

ANSWER:Test case 1 :

Input:

era ew sdneirf

Output:

are we friends

Test case 2 :

Input:

@olleh dlrow

Output:

Invalid Symbol Found

Test case 3 :

Input:

ym eman si dipa

Output:

my name is apid

Test case 4 :

Input:

nohtyp si nuf

Output:

python is fun

Test case 5 :

Input:

dlrow#olleh

Output:

Invalid Symbol Found

Test case 6 :

Input:

\$siht si ysae

Output:

Invalid Symbol Found